

SMART FERTILIZER SPRAYING SYSTEM USING PLANT DISEASE SEVERITY ANALYSIS

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Abstract—The growing need for precision agriculture calls for smart systems that can detect crop diseases and improve chemical use. Traditional fertilizer spraying methods often apply the same amount regardless of disease severity. This results in wasted resources and harm to the environment. To tackle this issue, this paper presents a Smart Fertilizer Spraying System that includes deep learning for classifying tomato disease severity. The system uses a Raspberry Pi with a USB camera and a YOLOv8 model built with the PyTorch framework. It analyzes images of tomato fruits and classifies them into three disease severity levels: low, medium, and high. Based on the predicted severity, control signals go to an Arduino microcontroller, which adjusts the fertilizer spraying time using relay-controlled pumps. The robotic platform allows for automated movement and operation. Experimental tests showed a model accuracy of about 75 to 80 percent under controlled conditions. This severity-based spraying method reduces fertilizer waste when compared to traditional uniform spraying and improves crop management precision. The system supports smarter automation in sustainable agriculture.

Index Terms—Precision Agriculture, YOLOv8, Deep Learning, Disease Severity Classification, Raspberry Pi, Arduino, Automated Fertilizer Spraying, Smart Farming

I. INTRODUCTION

Agriculture is crucial for global food security and economic growth, especially in developing countries where many people rely on farming for their livelihoods [1], [2]. However, crop productivity often suffers due to plant diseases, improper fertilizer use, and delayed infection detection, leading to significant yield losses and economic consequences [3], [4]. Traditional farming methods depend on manual inspections and conventional spraying techniques, which are slow, labor-intensive, and prone to human error [5]. Additionally, ex-

cessive or improper fertilizer application increases production costs and negatively impacts the environment, highlighting the need for smarter and more sustainable agricultural solutions [12].

Recent advancements in artificial intelligence, deep learning, and computer vision have enabled the development of precision agriculture systems capable of automated crop monitoring and disease detection [6], [11]. Convolutional Neural Networks (CNNs) and object detection models such as YOLO have demonstrated strong performance in real-time plant disease identification and localization [7]–[9]. These models provide faster detection speeds and improved accuracy compared to traditional image processing techniques.

Despite these technological advancements, many existing systems primarily focus on disease classification and provide advisory recommendations, still requiring manual intervention for treatment execution [10]. Furthermore, limited research has integrated disease severity estimation with automated fertilizer spraying mechanisms, resulting in inefficient and non-targeted chemical application.

This paper proposes a Smart Fertilizer Spraying System that integrates YOLOv8-based analysis of tomato disease severity. The system captures real-time images using a USB camera connected to a Raspberry Pi and performs severity classification using deep learning techniques. Based on the detected severity level—low, medium, or high—the system automatically adjusts the duration of fertilizer spraying through dedicated nozzles using an Arduino-controlled actuation mechanism. By combining real-time object detection, severity-based decision-making, and automated spraying con-

trol, the proposed system aims to optimize fertilizer usage, reduce human intervention, and promote sustainable precision agriculture.

II. LITERATURE SURVEY

Several research efforts have focused on improving agricultural productivity by combining image processing, machine learning, and automation technologies [1], [2]. These studies can be grouped into three main areas: traditional image-based disease detection systems, deep learning-based plant disease classification, and automated robotic spraying mechanisms.

Early research on plant disease detection mainly used classical image processing techniques such as color segmentation, thresholding, and feature extraction methods [1], [4]. In these systems, researchers pre-processed leaf images to remove noise, isolated diseased regions, and classified them using conventional machine learning algorithms such as Naive Bayes or Support Vector Machines [1]. Although these approaches demonstrated reasonable accuracy under controlled conditions, their performance was highly dependent on lighting conditions and manual feature engineering. Furthermore, most of these systems functioned as advisory tools, requiring farmers to interpret the results and manually apply fertilizers or pesticides [5].

With advancements in deep learning, Convolutional Neural Networks (CNNs) significantly improved the accuracy of plant disease detection [6], [9]. CNN models automatically extract hierarchical features from images, eliminating the need for manual feature selection. More recently, real-time object detection models such as YOLO have been utilized for disease localization and classification in agricultural environments [7], [8]. These models offer faster detection speeds and improved robustness under real-world conditions. However, many existing systems focus solely on disease identification without incorporating severity estimation or automated treatment mechanisms.

Simultaneously, several researchers have proposed robotic pesticide spraying systems to reduce human exposure to chemicals and enhance spraying precision [10]. These systems typically use image processing to detect infected regions and activate sprayers accordingly. While such robotic platforms improve spraying accuracy, most implementations concentrate on pesticide application rather than fertilizer management. Moreover, limited research has explored dynamically adjusting spray duration based on quantified disease severity levels.

Additionally, AI-based crop advisory platforms have been developed to support farmers in crop selection, disease diagnosis, and fertilizer recommendation processes [11]. Although these systems improve accessibility and decision-making, they lack real-time field-level automation and still depend on manual intervention for treatment execution. They also fail to integrate severity-based decision mechanisms with hardware actuation systems.

A comprehensive review of the existing literature indicates that current solutions either emphasize disease classification, advisory services, or automated spraying independently. Very

few studies integrate deep learning-based severity assessment with automated fertilizer spraying within a unified robotic framework. The absence of severity-controlled spray duration results in inefficient chemical utilization and reduced precision in crop management.

Therefore, a clear research gap exists in developing an integrated system that combines real-time deep learning-based disease severity classification with automated, dynamically controlled fertilizer spraying. Addressing this gap can significantly enhance precision agriculture by optimizing resource utilization, minimizing environmental impact, and reducing human intervention.

III. EXISTING SYSTEMS

Conventional farming methods mostly depend on manual crop inspection and uniform fertilizer application. Farmers visually check plant leaves to identify symptoms of diseases, pest damage, or nutrient deficiencies. Afterward, fertilizers or pesticides are applied using hand-operated or tractor-mounted sprayers. Although this method is widely practiced, it is labor-intensive, time-consuming, and highly dependent on human expertise. Manual inspection often leads to inaccurate disease assessment, resulting in either excessive or insufficient chemical application. Uniform spraying without considering infection severity contributes to fertilizer wastage, increased production costs, and environmental pollution.

To overcome these limitations, several image-based plant disease detection systems have been developed [1], [4]. Early approaches utilized traditional image processing techniques such as color segmentation, thresholding, and feature extraction, combined with classical machine learning algorithms for disease classification [1], [5]. While these systems improved detection accuracy under controlled environments, they mainly functioned as advisory tools. Farmers were still required to interpret the results and manually apply fertilizers or pesticides, limiting automation and practical efficiency [2].

Recent advancements in deep learning have introduced Convolutional Neural Networks (CNNs) and real-time object detection models for automated plant disease identification [6], [7]. These systems offer improved accuracy and faster detection capabilities compared to conventional approaches. However, most existing implementations focus primarily on disease classification or localization without estimating disease severity for decision-making. Furthermore, many systems lack integration with automated hardware control mechanisms, preventing real-time execution of severity-based treatment [3].

In summary, current systems either identify plant diseases without automation or implement spraying mechanisms without incorporating severity-based intelligence. The absence of an integrated framework that combines real-time deep learning-based severity classification with automated fertilizer spraying highlights a significant research gap. This limitation motivates the development of a precision agriculture system capable of performing severity-driven fertilizer control with minimal human intervention.

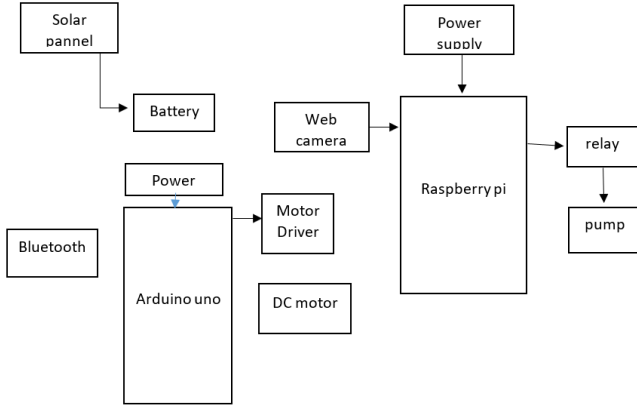


Fig. 1. Block diagram of the proposed smart fertilizer spraying system

IV. PROPOSED SYSTEM

The Smart Fertilizer Spraying System combines real-time disease detection with automated fertilizer spraying and robotic movement. The main goal of the system is to lessen manual effort, use fertilizer more efficiently, and enhance precision agriculture through smart decision-making.

A. System Architecture

The system's overall architecture includes four main parts: image acquisition, disease severity classification, actuation control, and power management.

In the image acquisition stage, a USB webcam captures real-time images of tomato crops in the field. These images are sent to the Raspberry Pi, which serves as the main processing unit. The Raspberry Pi runs a trained YOLOv8 deep learning model created with the PyTorch framework. The model examines the images and classifies any detected tomato disease into severity categories like low, medium, or high.

Unlike traditional systems that only identify the disease type, this system directly determines the severity level. This approach simplifies decision-making and enables immediate actions without needing extra processing.

After the severity level is identified, the Raspberry Pi sends the appropriate command to the Arduino Uno microcontroller via serial communication. The Arduino acts as the control and actuation unit. Based on the command received, it activates the right relay module connected to a fertilizer pump. Each severity level corresponds to a specific spraying duration, ensuring targeted fertilizer application.

This severity-based control system prevents over-fertilization in mildly affected crops and ensures adequate treatment in heavily infected areas. As a result, chemical waste is minimized, and crop treatment is more precise and effective.

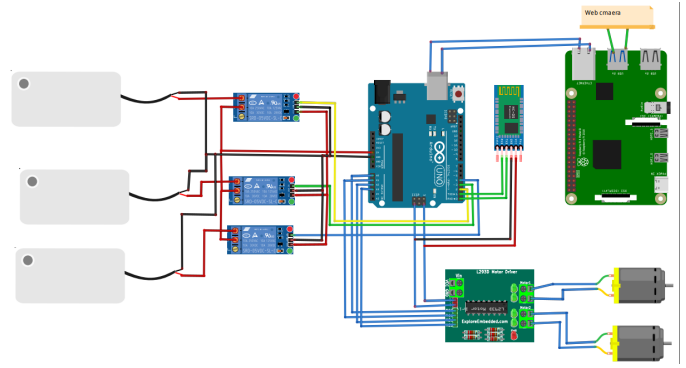


Fig. 2. Circuit Diagram of Proposed System

B. Hardware Implementation

The system's hardware includes a Raspberry Pi, Arduino Uno, relay modules, DC motors, a motor driver, water pumps, a battery, a solar panel, and a Bluetooth module.

The Arduino Uno controls the movement of the system. It drives the DC motors through an L293D motor driver, allowing the robot to move forward, backward, left, and right. This helps the system navigate the field efficiently.

Three relay modules connect to separate water pumps. These relays work in an active-high setup, meaning the pump activates when the Arduino sends a high signal. The pump activation duration varies based on the detected severity level. For example, low severity may trigger short spraying, medium severity may lead to moderate spraying, and high severity may require longer spraying.

Bluetooth communication is included for manual control and monitoring. This feature allows remote movement control and system supervision when needed.

For power management, the system runs on a 12V battery supported by a solar panel. The solar panel charges the battery during the day, ensuring continuous operation and reducing reliance on external power sources. This setup improves the sustainability and practicality of the system.

C. Component Description

The Raspberry Pi is the primary processing unit responsible for running the YOLOv8 model and performing severity classification. The Arduino Uno is the control unit that manages motor movement and pump activation. The motor driver controls the speed and direction of the DC motors. Relay modules act as switches for the high-power pumps. The DC motors provide movement for the robotic chassis. The water pumps dispense fertilizer. The battery and solar panel combination ensures continuous and efficient operation. The Bluetooth module allows for wireless communication and manual overrides.

D. System Advantages

The modular design of the system allows scalability and adjustments for different crop types and farming environments.

The severity-based spraying method ensures controlled chemical application, reducing excessive fertilizer use. Automated operation minimizes human exposure to chemicals and lessens the need for labor. Furthermore, integrating deep learning for decision-making improves consistency, reliability, and overall efficiency in fertilizer application.

By merging real-time object detection, severity classification, robotic movement, and automated spraying, the system provides a practical and intelligent solution for modern precision agriculture.

The hardware components used in the proposed system are summarized in figure 3.

TABLE I: Components Used in the smart fertilizer system

Component	Function	Purpose Description
Raspberry Pi	Main Processing Unit	Performs image processing, disease severity analysis and controls fertilizer spraying operations.
Arduino Uno	Control & Actuation Unit	Controls DC motors, water pumps, relays, and robot navigation based on commands from the Raspberry Pi.
Web Camera	Image Acquisition Unit	Captures real-time images of plant leaves for disease detection and severity analysis.
Motor Driver	Motor Control Interface	Controls the speed and direction of DC motors based on control signals from the Arduino.
DC Motor	Mobility Actuator	Provides movement to the robot chassis for field navigation.
DC Water Pump	Fertilizer Dispensing	Pumps fertilizer solution through spraying nozzles based on disease severity.
Relay Module	Switching Unit	Controls high-power devices such as water pumps using low-power control signals.
12V Battery	Power Management Unit	Provides continuous and energy-efficient power for system operation.
Bluetooth Module	Wireless Communication Unit	Enables wireless monitoring and basic control of the system.

Fig. 3. Components Used in the Smart Fertilizer System

V. METHODOLOGY

The proposed Smart Fertilizer Spraying System methodology consists of dataset preparation, severity labeling, deep learning model training, real-time inference, and hardware integration for automated fertilizer control.

A. Dataset Preparation

A total of 1,500 tomato images were used for training and evaluation of the model. The dataset consisted of three severity

classes: low, medium, and high, with 500 images in each class to ensure balanced class distribution. The images were manually collected and annotated.

The dataset was divided into training, validation, and testing subsets using a 70:20:10 ratio. Accordingly, 1,050 images were used for training, 300 images for validation, and 150 images for testing. This split ensured proper model generalization and reliable performance evaluation.

B. Severity Labeling Strategy

Disease severity was labeled based on the visible infection percentage on the tomato surface. The severity thresholds were defined as follows:

- Low severity: 0–25% infection
- Medium severity: 26–50% infection
- High severity: Greater than 50% infection

The infection percentage was estimated during dataset preparation, and each image was labeled according to its corresponding severity class. Thus, severity classification was incorporated directly into the training labels rather than being calculated dynamically during runtime.

C. YOLOv8n Model Training

The YOLOv8n (Nano) architecture from the Ultralytics framework was used for disease severity classification. The model was trained using the PyTorch deep learning framework.

Training was performed for 100 epochs with an input image size of 199×199 pixels. The dataset was provided in YOLO annotation format through a data configuration file. Default YOLOv8 hyperparameters were used during training. The model utilized the framework's default optimizer settings.

Training was conducted on a CPU-based system. The trained model weights were saved as *best.pt* and deployed for real-time inference on the Raspberry Pi.

D. Real-Time Inference Process

During real-time operation, a USB webcam connected to the Raspberry Pi captures live images of tomato crops. Each frame is processed by the trained YOLOv8n model for detection and classification.

The model directly predicts one of the predefined severity classes (low, medium, or high). No infection percentage is calculated during runtime. The system relies entirely on the model's classification output for decision-making.

A confidence threshold of 0.30 is applied during inference to filter low-confidence detections. The predicted severity label is transmitted to the Arduino microcontroller via USB serial communication.

E. Hardware Integration and Control Logic

The Raspberry Pi communicates with the Arduino Uno using USB serial communication. Based on the predicted severity class, the corresponding command string ("low", "medium", or "high") is sent to the Arduino.

Upon receiving the command, the Arduino activates the associated relay module connected to the fertilizer pump. Each

severity level is mapped to a predefined spraying duration as follows:

- Low severity: 2 seconds (2000 ms)
- Medium severity: 4 seconds (4000 ms)
- High severity: 6 seconds (6000 ms)

This time-based control mechanism ensures severity-dependent fertilizer application without additional computational overhead during runtime.

The integrated system enables coordinated operation between deep learning-based severity detection and hardware-based fertilizer spraying, ensuring efficient and automated precision agriculture.

VI. RESULTS AND DISCUSSION

A. RESULTS

We experimentally evaluated the proposed Smart Fertilizer Spraying System to assess its effectiveness in classifying disease severity and controlling fertilizer application. The YOLOv8n model was tested using a dataset containing 150 images divided into three severity classes: low, medium, and high.

The trained model achieved an overall classification accuracy of approximately 75–80% on the test dataset. Fig. 4 shows the training and validation accuracy obtained during the model training process.

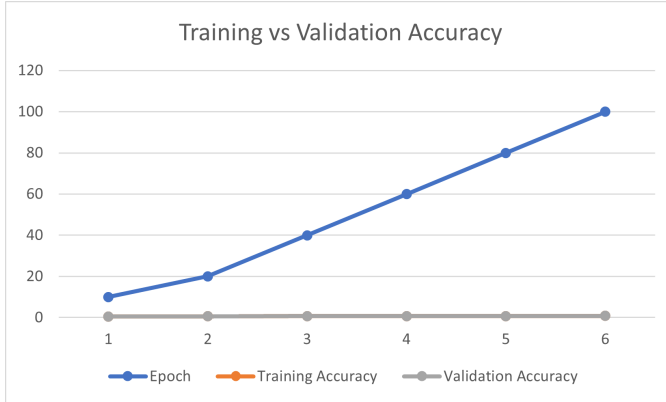


Fig. 4. Training and validation accuracy of the YOLOv8 model during the training process.

To further evaluate classification performance, a confusion matrix was generated as shown in Fig. 5. The diagonal elements represent correct classifications.

Actual / Predicted	Low	Medium	High
Low	42	5	3
Medium	6	38	6
High	4	7	39

Fig. 5. Confusion matrix of YOLOv8-based tomato disease severity classification.

Precision and recall values were also analyzed to evaluate the reliability of the classification model.

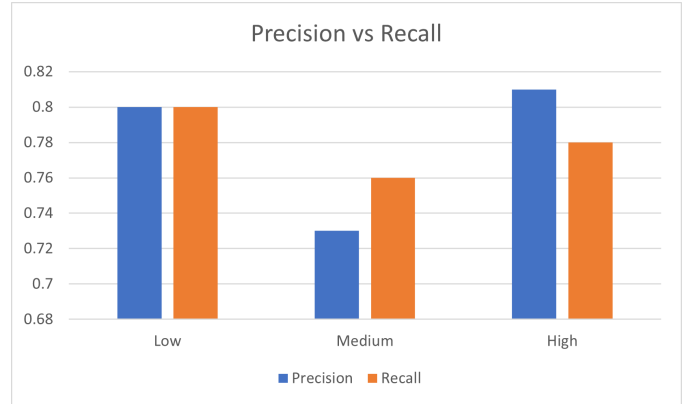


Fig. 6. Precision and recall comparison for tomato disease severity classes.

During real-time deployment, the Raspberry Pi processed webcam frames and transmitted the predicted severity label (“low”, “medium”, or “high”) to the Arduino Uno via USB serial communication. The spraying duration was adjusted according to the predicted severity level.

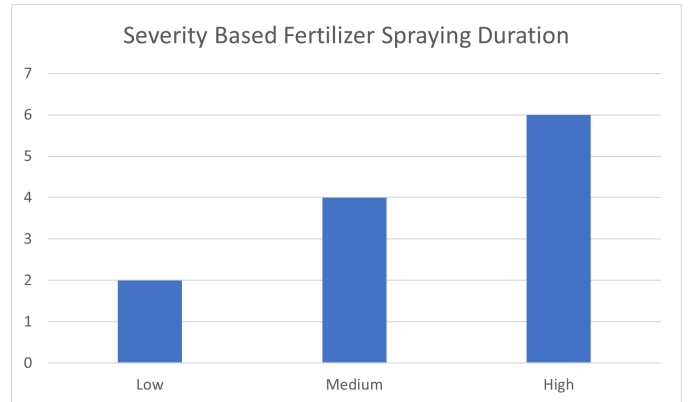


Fig. 7. Severity-based fertilizer spraying duration implemented in the proposed system.

Experimental tests confirmed that the relay modules and water pumps activated accurately based on the predicted severity

class. The robotic platform moved reliably while the spraying mechanism operated without noticeable communication delay between the Raspberry Pi and Arduino.

B. Discussion

The experimental results show that the YOLOv8n-based classification model can distinguish between different levels of tomato disease infection with moderate accuracy. An accuracy range of 75 to 80% suggests the system is good for practical precision agriculture, especially in controlled or semi-structured environments.

Unlike traditional uniform spraying systems, our approach allows for targeted fertilizer application. By linking predicted severity classes to specific spraying durations, we reduce chemical usage and promote efficient resource management.

The integration of the Raspberry Pi and Arduino creates reliable coordination between hardware and software. Since we predict severity directly as a class label, we do not need to calculate infection percentages at runtime. This reduces computational demands and simplifies decision-making.

However, there are some limitations. Our dataset was limited to 1,500 images, and we did not apply data augmentation techniques during training. Additionally, training was done on a CPU-based system, which might have limited optimization performance. Factors like lighting changes and camera positioning could also affect detection accuracy in open-field conditions.

Future improvements could include expanding the dataset, using data augmentation methods, conducting GPU-based training for better optimization, and testing system performance in various environmental settings. Extending the framework to analyze disease severity in multiple crops could further improve its use in precision agriculture.

VII. CONCLUSION

This work introduced a Smart Fertilizer Spraying System that combines deep learning-based disease severity classification with automated fertilizer control for precision agriculture. The main goal was to reduce manual work and improve fertilizer use based on severity-driven decision-making. By using a YOLOv8n model for real-time classification of tomato disease severity and connecting predicted severity levels to set spraying times, the system created an effective and smart fertilizer management method.

Experimental tests showed that the model achieved an overall classification accuracy between 75 to 80 percent, proving its ability to tell apart low, medium, and high severity levels in controlled settings. The reliable communication between the Raspberry Pi and Arduino allowed for effective operation of pumps and motors, which ensured synchronized robotic movement and precise fertilizer application. Adding solar power also improved the system's sustainability and practicality.

This study emphasizes the potential of integrating deep learning models with embedded hardware to create efficient precision agriculture solutions. By applying severity-based spraying instead of uniform chemical application, the proposed

system helps reduce fertilizer waste, enhance resource use, and promote environmentally friendly farming practices.

However, some limitations remain. These include a moderate dataset size, lack of data augmentation, and limitations from CPU-based training. Future work could focus on increasing the dataset size, adding advanced augmentation techniques, using GPU-based optimization, and testing the system in various real-world environments. Expanding the framework to support multi-crop disease severity detection could also boost its reach and effectiveness.

Overall, the proposed system shows that intelligent, severity-driven fertilizer control can be a practical step toward sustainable and technology-supported modern agriculture.

VIII. FUTURE SCOPE

The proposed system shows effective YOLOv8-based severity classification and automated fertilizer control. However, there are several ways to improve its performance, scalability, and practical use.

First, we can increase the dataset size to include more images taken in different environmental conditions, such as varying lighting, backgrounds, and crop growth stages. Using data augmentation techniques and conducting GPU-based training can further improve the model's generalization and detection accuracy.

Second, we can expand the system to support multi-crop disease severity detection by retraining the model with additional crop datasets. This would enhance flexibility and make the framework suitable for a wider range of agricultural applications.

From a hardware standpoint, adding GPS-based navigation and autonomous path planning algorithms can allow for fully autonomous field traversal. This would lessen manual intervention and increase coverage efficiency in large agricultural fields.

Future versions could also include IoT-based cloud connectivity for real-time data logging, remote monitoring, and long-term agricultural analysis. Storing historical severity data could help with predictive analysis and early disease outbreak detection.

Additionally, we can expand the framework to include pesticide spraying, nutrient deficiency detection, and soil condition monitoring modules. This would transform the system into a comprehensive precision agriculture platform.

By implementing these improvements, the proposed system can evolve into a scalable and smart farming solution that supports sustainable, data-driven agricultural practices.

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